Lemma 6.1.1: If $f \in \Lambda(X)$ is homogeneous of degree d, then f is determined by the terms of the coproduct $f[X_1+X_2]$ of degree k, $d-k$ in X_1, X_2 for $0<k<d$, together with the specialization $(\omega f)[1-q]$.

Proof: The terms of degree $k, d-k$ for $0<k<d$ in $f[X_1+X_2]$ determine f up to adding a primitive (* an element x of a Hopf algebra is primitive if $\Delta(x) = 1\otimes x + x\otimes 1$.) By "On the structure of Hopf algebra, 1965, Milnor & Moore), the power-sums $p_d(X)$ span the vector space of primitives in $\Lambda(X)$. Hence the result follows if the specialization of $p_d(X)$ does not vanish. Indeed, $(\omega p_d)[1-q] = (-1)^d(1-q^d) \neq 0$ and hence result follows. $\square$

Theorem: Let $\Phi = \Phi(z_1,\dots,z_\ell) \in \Gamma^\ell + I^\ell$ (not symmetric) be a rational function s.t. $\mathcal{P}(\Phi) = f[-MX^{m,n}]$, where $\mathcal{P}: S = \Gamma^+/I \longrightarrow \mathcal{E}^+$ is the shuffle to Schiffmann isomorphism in Theorem 3.2.1, and $f(X)$ is homogeneous of degree d, so $\ell = dm$. Assume that the denominator of Φ does not vanish when evaluated at any permutation of $(1,t,t^2,\dots,t^{\ell-1})$. Then

$$\Phi(1,t,\dots,t^{\ell-1}) = \frac{t^a(\omega f)[1-q]}{(1-q)^\ell} \quad \text{where } a = \tfrac{1}{2}d(dmn-m-n+1). \qquad (*)$$

Proof: We first show that $\Phi(1,t,\dots,t^{\ell-1})$ depends only on $\mathcal{P}(\Phi)$, or equivalently on

$$g(z) = \mathcal{P}_f(\Phi) = \sum_{\omega\in S_\ell} \omega\Big(\Phi(z)\prod_{\alpha\in R_+}((1-z^\alpha)(1-qz^{-\alpha})(1-tz^{-\alpha})(1-qtz^\alpha))\Big).$$

If $\omega \neq 1$, there is some index i s.t. $j = \omega^{-1}(i+1) < \omega^{-1}(i) = k$. Then the factor $\omega\left(1-t\frac{z_k}{z_j}\right) = 1-t\frac{z_i}{z_{i+1}}$ in $\omega\left(\prod_{\alpha\in R_+}(1-tz^\alpha)\right)$ vanishes at $z=(1,t,\dots,t^{\ell-1})$.

By our assumption on the denominator of Φ, we have

exists b/c $\Phi(1,\dots,t^{\ell-1})$ ha

$$\omega\Big(\Phi(z)\prod_{\alpha\in R_+}((1-z^\alpha)(1-qz^{-\alpha})(1-tz^{-\alpha})(1-qtz^\alpha))\Big) = 0 \quad \text{at } z=(1,t,\dots,t^{\ell-1}) \text{ for } \omega\neq 1.$$

Hence $g(1,t,\dots,t^{\ell-1}) = \Phi(1,t,\dots,t^{\ell-1})\prod_{1\le i\le j\le \ell}\left((1-t^{j-i})(1-qt^{j-i})(1-t^{j-i+1})(1-qt^{i-j+1})\right)$.

$\because$ $\prod_{1\le i\le j\le \ell}\left((1-t^{j-i})(1-qt^{j-i})(1-t^{j-i+1})(1-qt^{i-j+1})\right)$ is fixed and nonzero

$\therefore$ $g(1,t,\dots,t^{\ell-1})$ determines $\Phi(1,t,\dots,t^{\ell-1})$.

Next note that for mixed m,n,

$$a(d_1+d_2) = \tfrac{1}{2}(d_1+d_2)[(d_1+d_2)mn - m - n + 1]$$
$$= \tfrac{1}{2}d_1(d_1mn - m - n + 1) + \tfrac{1}{2}d_2(d_2mn - m - n + 1) + \tfrac{1}{2}(d_1d_2 + d_2d_1)mn = a(d_1) + a(d_2) + d_1d_2mn.$$

$a(d) = \frac{1}{2}d(dmn - m - n + 1)$ satisfies $a(d_1+d_2) = a(d_1) + a(d_2) + d_1d_2mn$.

Given d_1, d_2 s.t. $d_1+d_2 = d$, suppose $\Phi_1(z_1,\dots,z_k)$ and $\Phi_2(z_1,\dots,z_{\ell-k})$ satisfy $(*)$ with corresponding $f_1(X), f_2(X)$ homogeneous of degree d_1, d_2 respectively. Then $k = d_1m$, $\ell-k = d_2m = (d-d_1)m$, $\ell = dm$, Φ_1 and Φ_2 are homogeneous of degrees d_1n, d_2n respectively, and we have

$$\Phi_2(t^k,\dots,t^{\ell-1}) = (t^k)^{d_2n}\Phi_2(1,t,t^2,\dots,t^{\ell-k-1}) = t^{d_1m\cdot d_2n}\Phi_2(1,t,\dots,t^{\ell-k-1})$$

$$\therefore\ \Phi_1(1,t,\dots,t^{k-1})\Phi_2(t^k,\dots,t^{\ell-1}) = t^{d_1d_2mn}\Phi_1(1,t,\dots,t^{k-1})\Phi_2(1,t,\dots,t^{\ell-k-1}).$$

Recall that in $S = \Gamma^+/I$, the product is concatenation.

$$\therefore\ (\Phi_1\cdot\Phi_2)(1,t,\dots,t^\ell) = t^{d_1d_2mn}\cdot\frac{t^{a(d_1)}(\omega f_1)[1-q]}{(1-q)^k}\cdot\frac{t^{a(d_2)}(\omega f_2)[1-q]}{(1-q)^{\ell-k}} = t^{a(d_1+d_2)}\frac{(\omega(f_1f_2))[1-q]}{(1-q)^\ell}.$$

If $\Phi = \Phi_1\cdot\Phi_2$ with $f = f_1f_2$, then $(*)$ holds.

$\because$ $(*)$ is linear in Φ and f

$\therefore$ It suffices to check when $f = p_d$ which holds using Negut (The shuffle algebra revisited 2014). $\square$

i.e. $\Phi(1,t,\dots,t^{\ell-1}) = \frac{(-1)^{d-1}t^a(1-q^d)}{(1-q)^\ell}$ if $\mathcal{P}(\Phi) = p_d[-MX^{m,n}]$.

Corollary 6.1.3: Let $H = H(R_q, R_t, R_{qt}, \lambda)$ be an (m,n)-cuddly Catalanimal of length $\ell = dm$, and let $f \in \Lambda$ be homogeneous of degree d.

To show that $\mathrm{cub}(H) = f$ it suffices to verify the following:

(1) For $0 < k < d$, $f[X+Y]_{k,d-k} = \sum_{\substack{I \\ \sum_t |I_t| = km \\ |\lambda[I]_t| = kn}} (-1)^{|R_t^{I_t+I_t^c}|} (-q)^{+|R_q^{I_t^c-I_t}|} (-t)^{-|R_t^{I_t^c+I_t}|} (-qt)^{|R_{qt}^{I_t^c-I_t}|} \mathrm{cub}(H_I')(X)\, \mathrm{cub}(H_I'')(Y)$ where H_I', H_I'' are as in Theorem 5.3.1,

(2) $\Phi(z) = \Phi(R_q, R_t, R_{qt}, \lambda)$ satisfies $\Phi(1, t, \dots, t^{\ell-1}) = \dfrac{t^a (\omega f)[1-q]}{(1-q)^\ell}$ where $a = \frac{1}{2} d(dmn - m - n + 1)$.

Proof: By Lemma 6.1.1, f is determined by $f[X+Y]_{k,d-k}$ and $(\omega f)[1-q]$.

By Theorem 5.3.1, (1) gives the coproduct $\mathrm{cub}(H)(X+Y)$.

By Theorem 6.1.2, since $\Phi(z)$ has denominator $\prod_{\alpha \in R \setminus R_{qt}} (1 - qt z^\alpha)$ which does not vanish when $z = (1, t, \dots, t^{\ell-1})$, we have $\omega f[1-q] = \dfrac{(1-q)^\ell \cdot \Phi(1, t, \dots, t^{\ell-1})}{t^a}$.

Since Φ corresponds to H, $(\omega f)[1-q]$ is also "fixed" by H.

As a result, $f = \mathrm{cub}(H)$ if it satisfies both (1) and (2). $\square$